

Series Review 1 (covers 5.2 – 5.4)

Directions: Determine if each of the following series converges or diverges. Use any method.

1.
$$\sum_{k=1}^{\infty} \frac{1}{(3+k)^{2/5}}$$

2.
$$\sum_{k=1}^{\infty} \frac{\cos\left(\frac{1}{k}\right)}{k^2}$$

3. In the series $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$, x stands for a real number. Depending on what number x is replaced with the given series may converge or diverge. Write an argument to find the values of x for which $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$ converges.

$$4. \quad \sum_{k=1}^{\infty} \left(\frac{3}{2^k} - \frac{2}{3^k} \right)$$

$$5. \quad \sum_{k=1}^{\infty} \left(2 - \frac{1}{k} \right)$$

$$6. \quad \sum_{k=1}^{\infty} \frac{\arctan(k)}{1+k^2}$$